

Finite-index descent of confinement for compact orbit closures

Partial answer to Question 3.9 of arXiv:2604.18458

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Status. `candidate_partial_likely_valid`.

Abstract

Amrutam and Jiang ask whether a finite Pimsner–Popa-index subalgebra of a confined von Neumann subalgebra of a group factor must remain confined. We prove this when the conjugacy orbit closure of the larger algebra is compact. More sharply, any sequence which sends the smaller algebra to the scalars and for which the larger algebras have an Effros–Maréchal cluster point forces that cluster point to be finite-dimensional. In an i.c.c. group factor such a cluster point can itself be conjugated to the scalars. Thus every hypothetical counterexample to the unrestricted question must exhibit noncompact escape.

1 Source question

Let Γ be a countable i.c.c. group and put $A = L(\Gamma)$, with its canonical trace τ . For $\mathcal{B} \subseteq A$, write $E_{\mathcal{B}} : A \rightarrow \mathcal{B}$ for the trace-preserving conditional expectation and $\mathcal{B}^g = \lambda(g)\mathcal{B}\lambda(g)^*$.

Question 3.9 of Amrutam–Jiang [1] asks whether, for

$$\mathcal{N} \subseteq \mathcal{M} \subseteq A, \quad [\mathcal{M} : \mathcal{N}]_{\text{PP}} < \infty,$$

confinement of $\mathcal{M}$ implies confinement of $\mathcal{N}$. Here $\mathcal{B}$ is confined when $\mathbb{C}1$ is not in the Effros–Maréchal (EM) closure of $\{\mathcal{B}^g : g \in \Gamma\}$. The source page is reproduced in Figure 1.

2 The cluster-point theorem

Set

$$\lambda = [\mathcal{M} : \mathcal{N}]_{\text{PP}}^{-1} > 0.$$

Theorem 1 (Finite-index cluster obstruction). *Suppose $g_i \in \Gamma$ satisfies $\mathcal{N}^{g_i} \rightarrow \mathbb{C}1$ in the EM topology. If, along a subnet, $\mathcal{M}^{g_i} \rightarrow \mathcal{P}$, then $\mathcal{P}$ is finite-dimensional and every nonzero projection $e \in \mathcal{P}$ satisfies $\tau(e) \geq \lambda$. Moreover, $\mathcal{M}$ is not confined.*

Proof. For nested tracial subalgebras, $E_{\mathcal{N}}E_{\mathcal{M}} = E_{\mathcal{N}}$. The Pimsner–Popa inequality therefore gives, for every $x \in A_+$,

$$E_{\mathcal{N}}(x) = E_{\mathcal{N}}(E_{\mathcal{M}}(x)) \geq \lambda E_{\mathcal{M}}(x).$$

Conjugating yields

$$E_{\mathcal{N}^{g_i}}(x) \geq \lambda E_{\mathcal{M}^{g_i}}(x), \quad x \in A_+. \tag{1}$$

EM convergence is pointwise L^2 -convergence of conditional expectations. Since the positive cone is L^2 -closed, the limit of (1) is

$$\tau(x)1 \geq \lambda E_{\mathcal{P}}(x), \quad x \in A_+. \quad (2)$$

For $y \in \mathcal{P}_+$, take $x = y$ in (2); then $\tau(y)1 \geq \lambda y$. In particular, a nonzero projection $e \in \mathcal{P}$ satisfies $\tau(e) \geq \lambda$. A diffuse von Neumann algebra, or an infinite atomic one, has arbitrarily small nonzero projections. Thus $\mathcal{P}$ is finite-dimensional.

Lemma 1 below gives $\mathbb{C}1 \in \overline{\{\mathcal{P}^g : g \in \Gamma\}}^{\text{EM}}$. On the other hand, $\mathcal{P}$ belongs to the closed invariant orbit closure of $\mathcal{M}$, so the same is true of the entire orbit closure of $\mathcal{P}$. Hence $\mathbb{C}1$ belongs to the orbit closure of $\mathcal{M}$, and $\mathcal{M}$ is not confined. $\square$

Lemma 1 (Finite-dimensional algebras escape). *If Γ is i.c.c. and $\mathcal{P} \subseteq L(\Gamma)$ is finite-dimensional, then $\mathcal{P}$ is not confined.*

Proof. Choose an L^2 -orthonormal basis $1, a_1, \dots, a_d$ of $\mathcal{P}$, with $\tau(a_j) = 0$. Enumerate $\Gamma \setminus \{e\} = \{h_1, h_2, \dots\}$, and approximate every a_j in L^2 by finite Fourier sums.

For finite $F \subseteq \Gamma \setminus \{e\}$ and finite $S \subseteq \Gamma$, there is $g \in \Gamma$ such that $g^{-1}Fg \cap S = \emptyset$. Indeed, each forbidden equality $g^{-1}hg = s$ places g in a coset of $C_\Gamma(h)$; these centralizers have infinite index because Γ is i.c.c., and finitely many cosets of infinite-index subgroups cannot cover a group.

Applying this avoidance successively to the first n group elements and Fourier approximants of error $1/n$, choose g_n so that

$$\tau(a_j^* \lambda(g_n^{-1} h_k g_n)) \rightarrow 0 \quad (j = 1, \dots, d; k \geq 1).$$

The orthonormal-basis formula for conditional expectation then gives

$$\|E_{\mathcal{P}}(\lambda(g_n^{-1} h_k g_n))\|_2^2 = \sum_{j=1}^d |\tau(a_j^* \lambda(g_n^{-1} h_k g_n))|^2 \rightarrow 0.$$

Equivalently, $E_{\mathcal{P}^{g_n}}(\lambda(h_k)) \rightarrow 0$ for every $h_k \neq e$, while the expectations fix 1. Density of the group algebra and contractivity of conditional expectations imply $\mathcal{P}^{g_n} \rightarrow \mathbb{C}1$ in EM. $\square$

Corollary 1 (Compact orbit closure). *If the EM orbit closure of $\mathcal{M}$ is compact and $\mathcal{M}$ is confined, then $\mathcal{N}$ is confined.*

Proof. If $\mathcal{N}$ were not confined, choose $\mathcal{N}^{g_i} \rightarrow \mathbb{C}1$. Compactness supplies an EM-convergent subnet of $(\mathcal{M}^{g_i})$, contradicting Theorem 1. $\square$

In particular, confinement descends through finite index whenever $\mathcal{M}$ belongs to a compact uniformly recurrent subalgebra system. The theorem also gives a sharp necessary feature of any counterexample to the source question: every sequence sending $\mathcal{N}$ to the scalars must send $\mathcal{M}$ out of every EM-compact set.

3 Why the unrestricted upgrade stalls

The tempting inference from (1) is false without a cluster point. For the normalized trace on $M_d(\mathbb{C})$, scalar expectation satisfies

$$E_{\mathbb{C}1}(x) = \tau(x)1 \geq d^{-1}x \quad (x \geq 0),$$

but it annihilates every traceless matrix. Thus a finite-index quotient can hide precisely the non-scalar directions witnessing confinement.

The June 2026 follow-up [2] proves that subalgebra spaces of diffuse finite von Neumann algebras can contain closed copies of arbitrary Polish spaces and constructs noncompact uniformly recurrent systems. Consequently, there is no ambient compactness principle with which to manufacture the cluster point in Theorem 1. Finite module bases lead to moving coefficients after conjugation, and the positive-definite-function compactification does not preserve representability by subalgebras. Six focused upgrade attempts are documented in the accompanying attempt log.

Novelty and review. Searches using the exact question, “compact orbit closure”, “finite index”, “confined subalgebra”, and the June follow-up found no statement of Corollary 1 or Theorem 1. Novelty confidence is moderate because the argument is elementary and may be unindexed folklore. Expert review is recommended, especially for the EM limit passage and Lemma 1.

Limitation. This packet does not answer Question 3.9 in the genuinely noncompact case and does not claim a counterexample.

References

- [1] T. Amrutam and Y. Jiang, *C^* -simplicity, confined subalgebras, and operator algebraic uniform recurrence*, arXiv:2604.18458 (2026).
- [2] T. Amrutam, P. Fima, and Y. Jiang, *Uniformly recurrent subalgebras in finite von Neumann algebras*, arXiv:2606.09673 (2026).

We show that confinement reliably passes down to finite-index von Neumann subalgebras.

Lemma 3.8. *Let Γ be an infinite group. Let $\mathcal{M} \leq L(\Gamma)$ be a von Neumann subalgebra such that the Pimsner-Popa index $[L(\Gamma) : \mathcal{M}]$ is finite. Then $\mathcal{M}$ is a confined subalgebra in $L(\Gamma)$.*

Proof. According to [PP86], we know that there is a positive number $c > 0$ such that $\|E_{\mathcal{M}}(x)\|_2 \geq c\|x\|_2$ for all $0 \neq x \in L(\Gamma)_+$. Now assume that $\mathcal{M}$ is not confined, i.e., there is a sequence $s_i \in \Gamma$ such that $s_i \mathcal{M} s_i^{-1} \rightarrow \mathbb{C}$ under EM-topology. Then take any $0 \neq x \in L(\Gamma)_+$, we get that

$$c\|x\|_2 = c\|s_i^{-1} x s_i\|_2 \leq \|E_{\mathcal{M}}(s_i^{-1} x s_i)\|_2 \xrightarrow{i \rightarrow \infty} |\tau(x)|.$$

That is, we have shown that $|\tau(x)| \geq c\|x\|_2$ for all $0 \neq x \in L(\Gamma)_+$.

Since Γ is an infinite group, it is diffuse and thus contains a sequence of non-zero projections, say $\{p_i\}$, such that $\lim_{i \rightarrow \infty} \tau(p_i) = 0$. Then we deduce that $|\tau(p_i)| \geq c\|p_i\|_2 = c\sqrt{\tau(p_i)}$, i.e., $\tau(p_i) \geq c^2 > 0$. Taking $i \rightarrow \infty$ and since p_i goes to zero, we get that $0 \geq c^2 > 0$, a contradiction. $\square$

Question 3.9. *Let Γ be an i.c.c. group. Let $\mathcal{N} \leq \mathcal{M} \leq L(\Gamma)$ be von Neumann subalgebras such that the Pimsner-Popa index $[\mathcal{M} : \mathcal{N}]$ is finite. If $\mathcal{M}$ is a confined subalgebra in $L(\Gamma)$, then is $\mathcal{N}$ also confined?*

Figure 1: Source page 9: the finite-index setup and Question 3.9.